Anna Zheng - Reflection 1

In our first four classes of the module, we explored the fabrication and design of sensors and circuits, which served as my first introduction to the fields of materials and computer engineering. We started the first class by attempting to build a basic sensor using just graphite, water, and paper to understand how sensors utilize different inputs and outputs. From there, we toured several facilities over the next few days, including the maker space labs in the Wilbur Powerhouse, the MOCVD and cleanroom facilities in the Sinclair Laboratory, and the Whitaker Laboratory. During these sessions, we created interdigitated electrodes (IDE), utilizing both the precut ones we made with the vinyl cutter in Wilbur and homemade ones we cut from tape using an x-acto knife. We also learned how to deposit conductive materials like silver paste, silver paint, and carbon paint onto glass plates, using a hot plate to dry the metallic materials before measuring and comparing their resistivities. We concluded the technical portion in the fourth class by building a paper circuit in the shape of a butterfly using a battery, copper tape, and four LEDs. Finally, we spincoated a PEDOT-PSS polymer film onto our IDEs to observe how its resistivity changed when exposed to water vapor from our breath.

To accomplish all of this, my partner and I worked closely together through every step. Because neither of us had much prior knowledge in this field, we kind of unofficially decided to do everything together instead of delegating specific tasks, making it a shared learning experience. For our homemade IDEs, we both decided to cut simple rectangles, as they were easy to cut and allowed for easier calculations later on. Before testing our IDEs, we practiced finding the resistivity of a piece of copper tape. We used calipers to measure the dimensions of our materials and a multimeter to find the resistance. Using the equation $R = \rho L/A$, we calculated a resistivity of 8.88 x 10^-8 Ωm for our copper tape. Later, we applied this same process to our homemade IDEs, calculating a resistivity of 2.22 x 10^-5 Ωm for the silver paste, 2.66 x 10^-5 Ωm for the silver paint, and 4.84 x 10^-3 Ωm for the carbon paint. Calculating the resistivity for the carbon paint proved especially challenging because the material was so thin that the multimeter probes kept scraping it off the glass without taking a reading. We brainstormed solutions and added a small piece of copper tape to each end of the rectangle to serve as better contact points. While it wasn't a perfect fix and the resistance still fluctuated, we were able to record the lowest number to complete our calculations. For our PEDOT-PSS film, we didn't need to measure the thickness because it was predetermined by the spincoater's rpm. We calculated an ambient resistivity of 1.5 x 10^-3 Ωm, which experienced a 9.7% increase to 1.6 x 10^-3 Ωm after we breathed on it.

Through these activities, I learned a tremendous amount about the fields of material and computer engineering. Going into the class, I had no idea what materials engineers did, and I conflated computer engineering with software development. I learned that computer engineers actually design chips and sensors, as well as embedded systems. It is amazing to see how

materials scientists/engineers modify the properties of materials to create new functionalities that help improve our world. I also learned a lot about how physical conditions affect conductivity. For example, during the first class, my table threw out ideas like having sensors detect a chemical reaction or a change in current and resistance, and it was fun seeing how wet and dry graphite have different resistances. I had some prior knowledge of circuits coming in, and while most of what was gone over was helpful review, I was never taught how to add LEDs in circuits in my previous classes, so that was really cool. During our tours, I learned how to use a vinyl cutter by inputting material dimensions and uploading files. We also learned about a wide variety of other equipment available in the maker space labs, such as laser cutters, 3D printers, and CNC machines. Furthermore, Thomas Koenig and Renbo Song gave us a great introduction to photolithography and cleanroom equipment.

Overall, this initial introduction to engineering has been both an eye-opening and slightly scary experience. During the first class, I didn't really grasp the task and took a backseat while my tablemates brainstormed, which made me nervous about being surrounded by people who already knew what they were doing. However, it was incredibly reassuring to realize that my partner and many of my peers are in the same boat, and that Professor Winikus, Professor Strandwitz, and the upperclassmen are incredibly helpful and always ready to assist us. Jumping immediately into hands-on work was super fun, even when I faced challenges. My first IDE got messed up because I wasn't careful enough. For the paper circuit, I didn't really branch out creatively and based mine off the example provided. I ripped pieces of the copper tape before placing it on the template, and because I am sort of a perfectionist, combined with my long nails, it was a difficult and time-consuming process to place the tape down. In the end, only two of my LEDs lit up, but it was cool nonetheless. Seeing the MOCVD wires was intimidating, but putting on the special attire for the cleanroom felt surreal, and I hope to one day conduct research in those facilities. The resources in the maker space labs made me aware of what a privilege it is to be at an institution like Lehigh, and I definitely plan to use the vinyl cutter to make cute stickers soon. This module, thus far, has shown me how passionate my peers are, and it inspires me to continue working harder. I am excited about what I've been able to do so far and can't wait to learn even more.